

Optics coating — Solution

Y24 (2024) — English translation

A1

0.30 pts

Let a plane electromagnetic wave propagate in a substance with a refractive index n with an oscillation amplitude E of the electric field strength. Express the amplitude of oscillations of the magnetic field strength H in this wave in terms of E , n , as well as the magnetic and electric constants μ_0 and ε_0 , respectively.

In a plane electromagnetic wave, the volumetric energy densities of the electric and magnetic fields are the same, therefore:

$$w_M = \frac{\mu_0 \mu H^2}{2} = w_E = \frac{\varepsilon_0 \varepsilon E^2}{2}.$$

For the phase velocity of wave propagation we have:

$$v = \frac{c}{n} = \frac{1}{\sqrt{\mu_0 \varepsilon_0 \mu \varepsilon}} \Rightarrow n = \sqrt{\mu \varepsilon}.$$

Since $\mu = 1$, for H we have:

Answer:

$$H = nE \sqrt{\frac{\varepsilon_0}{\mu_0}}.$$

A2

0.40 pts

What 4 boundary conditions for the electric and magnetic fields are satisfied at the interface? Write them down on the answer sheets in a form convenient for you.

From Gauss's theorem for magnetic field induction it follows:

Since there are no free charges at the boundary, it follows from Gauss's theorem for the electric induction vector:

From the circulation theorem for the electric field strength it follows:

Since there are no external currents at the interface, from the circulation theorem for the magnetic field strength it follows:

Answer:

$$B_{n1} = B_{n2}.$$

A3

0.40 pts

All boundary conditions that are met at the interface between two media must be satisfied at each point at an arbitrary point in time. Based on this, prove the following equalities:

$$\omega_r = \omega_t = \omega \quad \theta_r = \theta \quad n_2 \sin \theta_t = n_1 \sin \theta.$$

Each of the fields in the medium 1 is the sum of the fields in the incident and reflected wave, and the field in the medium 2 is the field of the transmitted wave. Then each of the boundary conditions has the form:

$$Ae^{i(\omega t - \vec{k}_r \vec{r})} + Be^{i(\omega_r t - \vec{k}_r \vec{r})} = Ce^{i(\omega_t t - \vec{k}_t \vec{r})}.$$

The given condition must hold at any moment in time, from which follow relation:

$$\omega_r = \omega_t = \omega.$$

Also give condition must be satisfied in any point boundary between media. Let x – axis directed to the right on Fig. 1 and Fig. 2. Then for fulfillment boundary condition in any point boundary separation two media must hold relation:

$$k_{rx} = k_{tx} = k_x.$$

Wave number be determined relation:

$$k = \frac{\omega}{v} = \frac{\omega n}{c}.$$

Hence obtain:

$$k_x = \frac{\omega n_1 \sin \theta}{c} = k_{rx} = \frac{\omega n_1 \sin \theta_r}{c} = k_{tx} = \frac{\omega n_2 \sin \theta_t}{c}.$$

Hence follow relation:

$$\theta = \theta_r \quad n_1 \sin \theta = n_2 \sin \theta_t.$$

A4

0.50 pts

Derive an expression for the complex reflection and transmission coefficients r_s and t_s , respectively, in the case of a s -polarized wave. Express your answers using n_1 , n_2 , θ and θ_t . The expressions for r_s and t_s are called Fresnel's formulas for the s -polarized wave.

Let us write the boundary condition for the tangential component of the electric field strength:

$$E_{\tau 1} = E_0(1 + r_s) = E_{\tau 2} = E_0 t_s \Rightarrow 1 + r_s = t_s.$$

Write the boundary condition for the tangential component of the magnetic field intensity:

$$H_{\tau 1} = n_1 \sqrt{\varepsilon_0 / \mu_0} (1 - r_s) \cos \theta = H_{\tau 2} = n_2 \sqrt{\varepsilon_0 / \mu_0} t_s \cos \theta_t \Rightarrow n_1 (1 - r_s) \cos \theta = n_2 t_s \cos \theta_t.$$

From here we find:

Answer:

$$r_s = \frac{n_1 \cos \theta - n_2 \cos \theta_t}{n_1 \cos \theta + n_2 \cos \theta_t} \quad t_s = \frac{2n_1 \cos \theta}{n_1 \cos \theta + n_2 \cos \theta_t}.$$

A5

0.50 pts

Derive an expression for the coefficients r_p and t_p in the case of a p -polarized wave. Express your answers using n_1 , n_2 , θ and θ_t . The expressions for r_p and t_p are called Fresnel's formulas for the p -polarized wave.

Let us write the boundary condition for the tangential component of the electric field strength:

$$E_{\tau 1} = E_0(1 - r_p) \cos \theta = E_{\tau 2} = E_0 t_p \cos \theta_t \Rightarrow (1 - r_p) \cos \theta = t_p \cos \theta_t.$$

Write the boundary condition for the tangential component of the magnetic field intensity:

$$H_{\tau 1} = n_1 \sqrt{\varepsilon_0 / \mu_0} (1 + r_p) = H_{\tau 2} = n_2 \sqrt{\varepsilon_0 / \mu_0} t_p \Rightarrow n_1 (1 + r_p) = n_2 t_p.$$

From here we find:

$$r_p = \frac{n_2 \cos \theta - n_1 \cos \theta_t}{n_2 \cos \theta + n_1 \cos \theta_t} \quad t_p = \frac{2n_1 \cos \theta}{n_2 \cos \theta + n_1 \cos \theta_t}.$$

A6

0.30 pts

Determine the value of $\cos \theta_t$. Express your answer in terms of n_1 , n_2 and θ .

Let us introduce the axis z , directed vertically downwards. Then at an arbitrary point in space:

$$E_t \sim \exp \left(i \left(\omega t - \vec{k}_t \vec{r} \right) \right) = \exp \left(i \left(\omega t - k_{tx} x - k_{tz} z \right) \right).$$

Consider term $-ik_{tz}$:

$$-ik_{tz}z = -ik_t z \cos \theta_t.$$

At the same time for $\cos \theta_t$ have:

$$\cos \theta_t = \pm \sqrt{1 - \frac{n_1^2 \sin^2 \theta}{n_2^2}} = \pm \frac{i\sqrt{n_1^2 \sin^2 \theta - n_2^2}}{n_2}.$$

Damping amplitude wave correspond sign $-$, therefore:

Answer:

$$\cos \theta_t = -\frac{i\sqrt{n_1^2 \sin^2 \theta - n_2^2}}{n_2}.$$

A7

0.40 pts

The expressions for r_s and r_p can be represented as follows:

$$r_s = A_s e^{i\delta_p} \quad r_p = A_p e^{i\delta_p}.$$

Determine A_s , A_p , δ_s and δ_p . Answer express through n_1 , n_2 and θ .

Taking into account the expression for $\cos \theta_t$ under full reflection, the expressions for r_s and r_p take the form:

$$r_s = \frac{n_1 \cos \theta + i\sqrt{n_1^2 \sin^2 \theta - n_2^2}}{n_1^2 \cos \theta - i\sqrt{n_1^2 \sin^2 \theta - n_2^2}} \quad r_p = \frac{n_2 \cos \theta + \frac{in_1 \sqrt{n_1^2 \sin^2 \theta - n_2^2}}{n_2}}{n_2 \cos \theta - \frac{in_1 \sqrt{n_1^2 \sin^2 \theta - n_2^2}}{n_2}}.$$

The moduli of r_s and r_p equal unity due to the equality of the moduli of the real and imaginary parts of the numerator and denominator. Then they representation in trigonometric form look like following manner:

$$r_s = \exp \left(2i \arctg \left(\frac{\sqrt{\sin^2 \theta - (n_2/n_1)^2}}{\cos \theta} \right) \right).$$

$$r_p = \exp \left(2i \arctg \left(\frac{(n_1/n_2)^2 \sqrt{\sin^2 \theta - (n_2/n_1)^2}}{\cos \theta} \right) \right).$$

Thus:

Answer:

$$A_s = 1 \quad A_p = 1.$$

$$\delta_s = 2 \arctg \left(\frac{\sqrt{\sin^2 \theta - (n_2/n_1)^2}}{\cos \theta} \right) \quad \delta_p = 2 \arctg \left(\frac{(n_1/n_2)^2 \sqrt{\sin^2 \theta - (n_2/n_1)^2}}{\cos \theta} \right).$$

A8

0.40 pts

Define δ . Express your answer as one inverse trigonometric function in terms of θ and $n_{21} = n_2/n_1$.

Let's use the expression for the tangent of the difference:

$$\operatorname{tg}(\alpha - \beta) = \frac{\operatorname{tg} \alpha - \operatorname{tg} \beta}{1 + \operatorname{tg} \alpha \operatorname{tg} \beta}.$$

Hence:

$$\operatorname{tg}(\delta/2) = \frac{\operatorname{tg}(\delta_p/2) - \operatorname{tg}(\delta_s/2)}{1 + \operatorname{tg}(\delta_p/2) \operatorname{tg}(\delta_s/2)} = \frac{\sqrt{\sin^2 \theta - n_{21}^2}(n_{21}^{-2} - 1)}{\cos \theta \left(1 + \frac{(\sin^2 \theta - n_{21}^2)n_{21}^{-2}}{\cos^2 \theta} \right)}.$$

Transform obtain expression:

$$\operatorname{tg}(\delta/2) = \frac{\cos \theta \sqrt{\sin^2 \theta - n_{21}^2} (n_{21}^{-2} - 1)}{\sin^2 \theta (n_{21}^{-2} - 1)} = \frac{\cos \theta \sqrt{\sin^2 \theta - n_{21}^2}}{\sin^2 \theta}.$$

Thus:

Answer:

$$\delta = 2 \operatorname{arctg} \left(\frac{\cos \theta \sqrt{\sin^2 \theta - n_{21}^2}}{\sin^2 \theta} \right).$$

B1

0.10 pts

At what values of the acute angle α of the Fresnel rhombus will the electromagnetic wave experience total reflection from the faces of the Fresnel rhombus BC and AD ? Express your answer in terms of n and calculate it in degrees accurate to four significant figures.

The angle of incidence on the faces BC and AD will be equal to α , therefore the condition must be satisfied:

$$n \sin \alpha \leq 1.$$

Thus:

Answer:

$$\alpha \in [\arcsin(1/n); \pi/2] = [41.47^\circ; 90^\circ).$$

B2

0.40 pts

At what values of the angle $\alpha = \alpha'$ can we obtain circular polarization from linear polarization using the Fresnel rhombus? Express your answers in terms of n and calculate them in degrees to four significant figures.

At each total reflection, a phase shift of $\pi/4$ occurs between the p -polarization and the s -polarization, therefore:

$$\operatorname{tg}(\delta/2) = \operatorname{tg}(\pi/8) = \sqrt{\frac{1 - \cos \delta}{1 + \cos \delta}} = \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2} + 1}} = \sqrt{2} - 1.$$

Thus:

$$\sqrt{2} - 1 = \frac{\cos \alpha' \sqrt{\sin^2 \alpha' - (1/n)^2}}{\sin^2 \alpha'}.$$

Hence:

$$\sin^4 \alpha' (4 - 2\sqrt{2}) - \sin^2 \alpha' (1 + (1/n)^2) + (1/n)^2 = 0$$

Solve square equation, obtain:

$$\alpha' = \arcsin \sqrt{\frac{1 + (1/n)^2 \pm \sqrt{(1 + (1/n)^2)^2 - 4(1/n)^2(4 - 2\sqrt{2})}}{8 - 4\sqrt{2}}} \approx 48.62^\circ, 54.62^\circ.$$

Note that both angles satisfy the condition obtained in part B1 and are suitable.

Answer:

$$\alpha' = \arcsin \sqrt{\frac{1 + (1/n)^2 \pm \sqrt{(1 + (1/n)^2)^2 - 4(1/n)^2(4 - 2\sqrt{2})}}{8 - 4\sqrt{2}}} \approx 48.62^\circ, 54.62^\circ.$$

B3

0.30 pts

Let a beam of light fall on a Fresnel rhomb over the entire area of face AB . Angle $\alpha = \alpha'$. At what values of BC/AB does the Fresnel rhomb reverse the entire beam of rays (all rays entering the Fresnel rhomb through face AB experience two total reflections)? Calculate your answer to three significant figures.

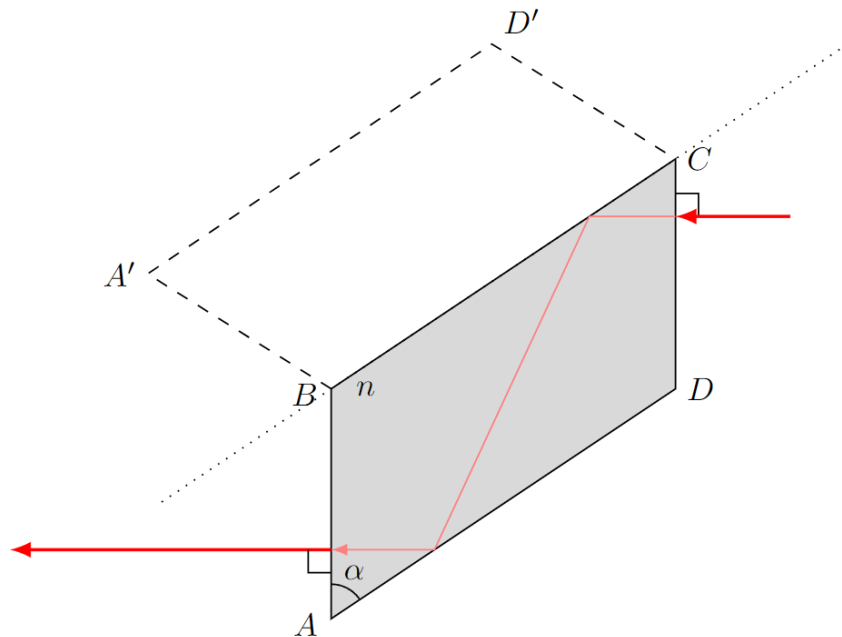
Let's use the sweep method. The Fresnel rhomb reverses the entire beam of rays subject to:

$$BC \cos \alpha + AB \cos 2\alpha - AB = 0 \Rightarrow \frac{BC}{AB} = \frac{2 \sin^2 \alpha}{\cos \alpha}.$$

Thus:

Answer:

$$\frac{BC}{AB} = \begin{cases} 1.70 & \text{at } \alpha' = 48.62^\circ \\ 2.30 & \text{at } \alpha' = 54.62^\circ \end{cases}$$



B4

0.10 pts

At what values of the angle α of the Mooney rhomb will the electromagnetic wave experience total reflection from the faces BC and CD of the Fresnel rhomb? Express your answer in terms of n and calculate it in degrees accurate to four significant figures.

The angle of incidence on the faces BC and CD will be equal to α , therefore the condition must be satisfied:

$$n \sin \alpha \leq 1.$$

Thus:

Answer:

$$\alpha \in [\arcsin(1/n); \pi/2) = [41.47^\circ; 90^\circ).$$

B5

0.30 pts

At what values of the angle $\alpha = \alpha'$ can we obtain circular polarization from linear polarization using the Mooney rhomb? Express your answers in terms of n and calculate them in degrees to four significant figures.

Since the number of reflections is two, the answer for this item is the same as the answer for item B2:

Answer:

$$\alpha' = \arcsin \sqrt{\frac{1 + (1/n)^2 \pm \sqrt{(1 + (1/n)^2)^2 - 4(1/n)^2(4 - 2\sqrt{2})}}{8 - 4\sqrt{2}}} \approx 48.62^\circ, 54.62^\circ.$$

B6

0.30 pts

Let the angle be $\alpha = \alpha'$. Determine the maximum possible value γ of the ratio of the cross-sectional area of the beam incident on face AB to the area of face AB , if the Mooney rhombus reverses the entire beam of rays (all rays entering the Mooney rhombus through face AB experience two total reflections). Calculate your answer to three significant figures.

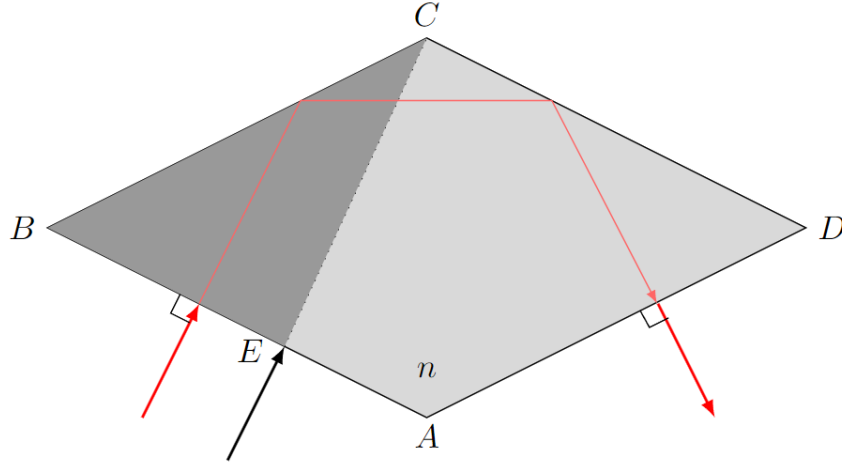
The rays experience two total reflections if they first hit the BC face. Let E be a point on face AB , after passing which the ray hits point C . Then the position of the point E is determined by the relation:

$$BE/BC = BE/AB = \cos \alpha.$$

For record value α maximum ratio area transverse cross-section beam to area face AB equal ratio $BE/AB = \cos \alpha$. We have:

Answer:

$$\gamma_{max} = \begin{cases} 0.661 & \text{at } \alpha' = 48.62^\circ \\ 0.579 & \text{at } \alpha' = 54.62^\circ \end{cases}$$



C1

0.30 pts

Show the validity of the following relations:

$$r'_1 = -r_1 \quad 1 - r_1^2 = t_1 t'_1.$$

Let ψ be the angle of refraction of the incident wave. Then ψ is equal to the angle of incidence of the wave with wave vector $\vec{k}'$ on the interface between media with refractive indices n_1 and n , and the angle of refraction is equal to θ . In the case of s -polarization we have:

$$r_{1s} = \frac{n_1 \cos \theta - n \cos \psi}{n_1 \cos \theta + n \cos \psi} \quad r'_{1s} = \frac{n \cos \psi - n_1 \cos \theta}{n_1 \cos \theta + n \cos \psi}.$$

In the case of p -polarization have:

$$r_{1p} = \frac{n \cos \theta - n_1 \cos \psi}{n \cos \theta + n_1 \cos \psi} \quad r'_{1p} = \frac{n_1 \cos \psi - n \cos \theta}{n \cos \theta + n_1 \cos \psi}.$$

Thus, in both cases the equality $r'_1 = -r_1$ is satisfied. Let us determine t'_1 in the case of both polarization:

$$t'_{1s} = \frac{2n \cos \psi}{n_1 \cos \theta + n \cos \psi} \quad t'_{1p} = \frac{2n \cos \psi}{n_1 \cos \psi + n \cos \theta}.$$

At the same time for t_{1s} and t_{1p} have:

$$t_{1s} = \frac{2n_1 \cos \theta}{n_1 \cos \theta + n \cos \psi} \quad t_{1p} = \frac{2n_1 \cos \theta}{n \cos \theta + n_1 \cos \psi}.$$

Hence:

$$1 - r_{1s}^2 = \frac{4nn_1 \cos \theta \cos \psi}{(n_1 \cos \theta + n \cos \psi)^2} = \frac{2n_1 \cos \theta}{n_1 \cos \theta + n \cos \psi} \cdot \frac{2n \cos \psi}{n_1 \cos \theta + n \cos \psi} = t_{1s} t'_{1s}.$$

$$1 - r_{1p}^2 = \frac{4nn_1 \cos \theta \cos \psi}{(n \cos \theta + n_1 \cos \psi)^2} = \frac{2n_1 \cos \theta}{n \cos \theta + n_1 \cos \psi} \cdot \frac{2n \cos \psi}{n \cos \theta + n_1 \cos \psi} = t_{1p} t'_{1p}.$$

Thus, equality $1 - r_1^2 = t_1 t'_1$ is completed.

C2

0.60 pts

Express E_r and E in terms of E_1 , E' , r_1 , r'_1 , t_1 and t'_1 .

If there were no wave with electric field strength E' , then the complex amplitudes $E_r^{(1)}$ and $E^{(1)}$ would be as follows:

$$E_r^{(1)} = r_1 E_1 \quad E^{(1)} = t_1 E_1.$$

If a wave with electric-field intensity E_1 be absent, that complex amplitude $E_r^{(2)}$ and $E^{(2)}$ be the following:

$$E_r^{(2)} = r'_1 E' \quad E^{(2)} = t'_1 E'.$$

The resulting complex amplitudes of electric field strengths are found from the superposition principle:

Answer:

$$E_r = r_1 E_1 + t'_1 E' \quad E = t_1 E_1 + r'_1 E'.$$

C3

0.60 pts

Reasoning similarly to C2, express E' and E_t in terms of E , r_2 and t'_2 , n_1 , n_2 , n , k , d and θ . Note that the complex amplitudes of the electric fields at point O' are out of phase with respect to the corresponding complex amplitudes at point O .

Since point O' is shifted relative to point O , the complex amplitudes $E_{O'}$, $E'_{O'}$ and $E_{O't}$ of the electric field strengths in waves with wave vectors $\vec{k}'$, $\vec{k}$ and $\vec{k}_t$ are shifted in phase relative to E , E' and E_t . Considering a wave with wave vector $\vec{k}'$ as a reflected wave with wave vector $\vec{k}$, we obtain:

$$r_2 E e^{-ikd \cos \psi} = E' e^{ikd \cos \psi}.$$

Similarly consider wave with wave vector $\vec{k}_t$ as the transmitted wave with wave vector $\vec{k}$, obtain:

$$t_2 E e^{-ikd \cos \psi} = E_t e^{-ik_t d \cos \varphi},$$

where φ – refraction angle wave with wave vector $\vec{k}$. Write expression for $\cos \varphi$ and $\cos \psi$:

$$\cos \psi = \frac{\sqrt{n^2 - n_1^2 \sin^2 \theta}}{n} \quad \cos \varphi = \frac{\sqrt{n_2^2 - n_1^2 \sin^2 \theta}}{n_2}.$$

Account for also relation:

$$k_t = \frac{kn_2}{n}.$$

Finally:

Answer:

$$E' = r_2 E \exp \left(-\frac{2ikd\sqrt{n^2 - n_1^2 \sin^2 \theta}}{n} \right)$$

$$E_t = t_2 E \exp \left(\frac{ikd \left(\sqrt{n_2^2 - n_1^2 \sin^2 \theta} - \sqrt{n^2 - n_1^2 \sin^2 \theta} \right)}{n} \right).$$

C4

0.60 pts

Using the results from steps C1-C3, determine the reflectance $r = E_r/E_1$. Express your answer in terms of r_1 , r_2 , n_1 , n , k , d and θ .

We will temporarily write the expression for E' through E in the following form:

$$E' = r_2 E e^{-i\Delta}.$$

Express E through E_1 :

$$E = t_1 E_1 + r_1' E' = t_1 E_1 + E r_1' r_2 e^{-i\Delta} \Rightarrow E = \frac{t_1 E_1}{1 - r_1' r_2 e^{-i\Delta}} = \frac{t_1 E_1}{1 + r_1 r_2 e^{-i\Delta}}.$$

Use expression for E_r :

$$E_r = r_1 E_1 + t_1' E' = r_1 E_1 + \frac{E_1 t_1 t_1' r_2 e^{-i\Delta}}{1 + r_1 r_2 e^{-i\Delta}} = E_1 \left(r_1 + \frac{r_2 (1 - r_1^2) e^{-i\Delta}}{1 + r_1 r_2 e^{-i\Delta}} \right).$$

Thus:

$$r = \frac{r_1 + r_2 e^{-i\Delta}}{1 + r_1 r_2 e^{-i\Delta}}$$

Substitute Δ , we find:

Answer:

$$r = \frac{r_1 + r_2 \exp \left(-\frac{2ikd\sqrt{n^2 - n_1^2 \sin^2 \theta}}{n} \right)}{1 + r_1 r_2 \exp \left(-\frac{2ikd\sqrt{n^2 - n_1^2 \sin^2 \theta}}{n} \right)}.$$

C5

0.70 pts

At what value of the plate thickness d is it possible for the reflection coefficient to vanish? Express your answer in terms of λ_1 , n_1 and n .

At normal incidence, the s -polarization and p -polarization are equivalent to each other, so we will consider only the s -polarized wave. The reflection coefficients r_1 and r_2 are real, so for the reflection coefficient to be zero, the quantity $e^{-i\Delta}$ must be real. This is possible with:

$$2kd = \pi N \Rightarrow d = \frac{\lambda N}{4}.$$

Relate quantity λ and λ_1 :

$$\lambda = vT = \frac{2\pi c}{\omega n} \Rightarrow \lambda_1 n_1 = \lambda n.$$

Answer: Thus:

$$d = \frac{\lambda_1 n_1 N}{4n}.$$

C6

0.80 pts

At what values of the refractive index of the plate n is it possible for the reflection coefficient to vanish? Express your answer in terms of n_1 and n_2 .

Let us analyze the case of odd N :

$$r_1 - r_2 = 0 \Rightarrow r_1 = \frac{n_1 - n}{n_1 + n} = r_2 = \frac{n - n_2}{n + n_2},$$

from which:

$$n_1 n + n_1 n_2 - n^2 - n_2 n = n_1 n - n_1 n_2 + n^2 - n n_2 \Rightarrow n = \sqrt{n_1 n_2}$$

Let us analyze the case of even N :

$$r_1 + r_2 = 0 \Rightarrow n_1 n + n_1 n_2 - n^2 - n_2 n = -n_1 n + n_1 n_2 - n^2 + n n_2 \Rightarrow n = 0.$$

Thus, possible only case of odd N , and in fact quantity d can take only following value:

$$d = \frac{\lambda(1 + 2m)}{4} \sqrt{\frac{n_1}{n_2}}.$$

Finally:

Answer:

$$n = \sqrt{n_1 n_2}$$

D1

0.20 pts

What is the refractive index n' of the antireflection coating? Express your answer using n .

Let's use the result of point C6:

Answer:

$$n' = \sqrt{n}.$$

D2

0.50 pts

What is the focal length of the lens f ? Express your answer using R and n

Let us obtain an expression for the focal length of a biconvex lens with radii of curvature of the surface with radii of curvature R_1 and R_2 , made of a substance with a refractive index n . Let's use the principle of tautochrone in the paraxial approximation:

$$l_{\text{opt}} = nh + f = \frac{r^2}{2R_1} + n \left(h - \frac{r^2}{2R_1} - \frac{r^2}{2R_2} \right) + \frac{r^2}{2R_2} + f + \frac{r^2}{2f}.$$

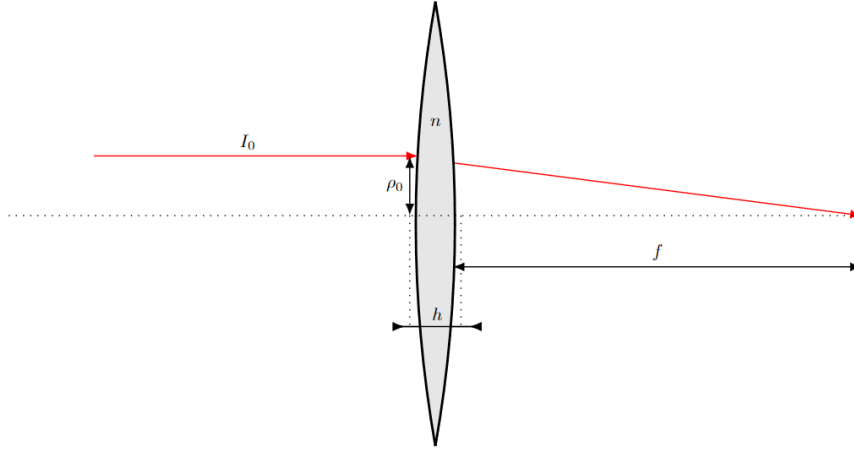
From the independence of r we obtain:

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right).$$

Since $R_1 = R_2 = R$:

Answer:

$$f = \frac{R}{2(n - 1)}.$$



D3

1.00 pts

Determine the force F acting on the lens. Express your answer in terms of I_0 , r , R , n and c .

The emergence of a force acting on the lens is due to the rotation of the wave front after passing through the lens. Let p_{x0} and p_x be the projections of the photon momentum onto the x axis before and after passing through the lens. Then the momentum Δp_x transferred by the photon to the lens is:

$$\Delta p_x = p_{x0} - p_x.$$

Let a wave falls on the lens at a distance ρ from its axis. After passing through the lens, the wave front rotate on angle $\varphi \approx \rho/f$, therefore have:

$$\Delta p_x(\rho) = p_{x0}(1 - \cos \varphi) \approx \frac{p_{x0}\rho^2}{2f^2}.$$

Let n – concentration photon in fall on lens beam, and E – energy one photon. Then force, act ring with internal and external radius ρ and $\rho + d\rho$ constitute:

$$dF_x = n_0 c \cdot 2\pi \rho d\rho \cdot \Delta p_x(\rho) = \frac{\pi n_0 p_{x0} c}{f^2} \cdot \rho^3 d\rho = \frac{\pi n_0 E}{f^2} \cdot \rho^3 d\rho.$$

At the same time $I_0 = n_0 c E$, therefore:

$$F_x = \frac{\pi I_0}{f^2 c} \int_0^r \rho^3 d\rho = \frac{\pi I_0 r^4}{4f^2 c}.$$

Substitute f , we get:

Answer:

$$F = \frac{\pi r^4 I_0 (n - 1)^2}{R^2 c}.$$